Symmetric Noise Stability on the Hypercube: Near Half-Order

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Abstract

For balanced Boolean functions on the discrete hypercube, we study the conjecture that dictator functions maximize the symmetric noise stability of order $1/2$ (equivalently, maximize conditional Tsallis or Rényi entropy of order $1/2$). We prove that the conjecture holds for $0 \le \rho \le 1971/2000 = 0.9855$, as well as, for every order α in $13/25 \le \alpha < 1$ and every $0 \le \rho \le 1$.

Keywords. Noise stability; Boolean functions; noise operators; Tsallis entropy; Rényi entropy.
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1 Introduction

Let $X = (X_1, \dots, X_n)$ be uniformly distributed on $\{-1, 1\}^n$. For $0 \le \rho \le 1$, let Y be obtained from X by independently flipping each coordinate with probability $p = (1 - \rho)/2$. The Bonami–Beckner noise operator is

$$T_\rho u(x) = \mathbb{E}[u(Y) \mid X = x].$$

If $f : \{-1, 1\}^n \to \{0, 1\}$ is balanced, write $g = 2f - 1$. The conditional probability of $f(Y) = 1$ given $X = x$ is $T_\rho f(x)$. Set

$$J(u) = \sqrt{\frac{1+u}{2}} + \sqrt{\frac{1-u}{2}}.$$

For a moment parameter $\alpha \ne 1$, the symmetric α-Tsallis function used in [1] is

$$\Phi_\alpha^{\mathrm{sym}}(z) = \frac{z^\alpha + (1-z)^\alpha - 1}{\alpha - 1}.$$

At $\alpha = 1/2$, this equals $-2(J(2z-1)-1)$. Maximizing the symmetric $1/2$-Tsallis stability $\mathbb{E}\Phi_{1/2}^{\mathrm{sym}}(T_\rho f)$ therefore means minimizing the "square-root functional" $\mathbb{E}J(T_\rho g)$.

Our central problem is

Conjecture 1 (Symmetric $1/2$-stability [1])**.** For every balanced Boolean f and every $\rho \in [0, 1]$,

$$\mathbb{E}J(T_\rho g) \ge J(\rho). \tag{1}$$

That is, a coordinate indicator $f(x) = \mathbf{1}_{\{x_i=1\}}$ or $f(x) = \mathbf{1}_{\{x_i=-1\}}$ (known as a dictator function) maximizes the symmetric $1/2$-stability.

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The problem is a special case of the symmetric Φ-stability program [2]. It is related to, but distinct from, the Shannon-information extremum proposed by Courtade and Kumar [3], the Tsallis-information extremum of order in $(1, 2)$ proposed by Li and Médard [4], as well as the Hellinger-information extremum of Anantharam and collaborators [5]. The Gaussian-profile approach of Chen and Nair [8], and the sharp critical isoperimetric inequality of Durcik–Ivanisvili–Roos–Xie [7] provide complementary points of reference. The conjecture 1 strengthens or generalizes the balanced version of these conjectures.

The dimension-induction component follows the local-to-product philosophy of Bobkov [9], with auxiliary profiles tailored to the square-root functional. Fourier conventions are as in [10]; the Gaussian invariance theory [11] gives related perspective but is not a premise of our certification. Recent manuscripts on the Shannon most-informative Boolean-function conjecture [12, 13, 14, 15, 16] concern a different entropy and do not directly imply our noise stability comparisons.

We establish the following three main statements.

Theorem 1 ($1/2$-stability for $\rho \le 0.9855$)**.** *For every $n \ge 1$ and every balanced $g : \{-1, 1\}^n \to \{-1, 1\}$,*

$$\mathbb{E}J(T_\rho g) \ge J(\rho), \qquad 0 \le \rho \le \rho_{\text{cert}} := \frac{1971}{2000} = 0.9855. \tag{2}$$

One may further improve the upper bound 0.9855 by optimizing the kernel in (10). However, we do not believe using the kernel of the form given in (10) can improve this bound to 1. In other words, it requires a new form of kernels.

For correlation above 0.9855, we in fact have shown a local optimality theorem, which says that for every balanced Boolean g with $\delta(g) \le 1/10$,

$$\mathbb{E}J(T_\rho g) - J(\rho) \ge \frac{\delta(g)}{1000}\sqrt{1-\rho^2}, \qquad \frac{49}{50} \le \rho \le 1, \tag{3}$$

where the distance $\delta(g)$ from a signed dictator is defined by

$$\delta(g) = \min_{1\le i\le n,\, \sigma\in\{-1,1\}} \mathbb{P}(g(X) \ne \sigma X_i) = \frac{1 - \max_i |\widehat{g}(\{i\})|}{2}.$$

For the power comparison put

$$J_\alpha(u) = \left(\frac{1+u}{2}\right)^\alpha + \left(\frac{1-u}{2}\right)^\alpha.$$

Theorem 2 (α-stability for $\alpha \ge 0.52$)**.** *For every balanced Boolean $g : \{-1, 1\}^n \to \{-1, 1\}$, every $0.52 = 13/25 \le \alpha < 1$, and every $0 \le \rho \le 1$,*

$$\mathbb{E}J_\alpha(T_\rho g) \ge J_\alpha(\rho). \tag{4}$$

It is already known [17, 1] that for $0 < a < b < 1$, if dictators maximize symmetric a-Tsallis stability, then dictators maximize symmetric b-Tsallis stability. Hence, this theorem resolves a weaker version of Conjecture 1.

Proof structure. Sections 3 gives the common tool—the dimension induction. Sections 4–5 apply them to the two theorems. The appendices collect the semigroup argument, shared certificate methods, and exact parameter data. The verification reproducibility are recorded in Appendix D.

2 Notation and preliminary reductions

All cube expectations use normalized uniform measure. We use $f = 1_A \in \{0,1\}$ and $g = 2f - 1 \in \{-1,1\}$, and distinguish the crossover probability $p = (1-\rho)/2$ from $s = \sqrt{1-\rho^2} = 2\sqrt{p(1-p)}$. Write $q = 1 - p$. Set

$$F_\alpha(z) = z^\alpha + (1-z)^\alpha - 1, \quad F = F_{1/2}, \qquad c_s = F(p) = \sqrt{1+s} - 1, \qquad C_* := \sqrt{2} - 1. \tag{5}$$

Thus $F(z) = F(1-z)$, $F \geq 0$, and F is strictly concave in the interior. Linearity and preservation of constants give

$$T_\rho g = 2T_\rho f - 1, \qquad J(T_\rho g) = \sqrt{T_\rho f} + \sqrt{1 - T_\rho f}, \qquad J(u) = \sqrt{1 + H(u)}, \tag{6}$$

where

$$H(u) = \sqrt{1-u^2}.$$

For a dictator $f_d = 1_{\{x_i=1\}}$, the two noise outputs are p, q. Hence its raw square-root functional is $\sqrt{p} + \sqrt{q} = 1 + c_s$. At $\rho = 0$, balance gives $T_0 f = 1/2$ and the comparison is an identity. At $\rho = 1$, Booleanity makes both sides equal to 1. Thus only interior correlations require proof.

2.1 Fourier and semigroup conventions

For $S \subseteq [n]$, let $\chi_S(x) = \prod_{i \in S} x_i$. Then

$$\widehat{g}(S) = \mathbb{E} g \chi_S, \qquad g = \sum_S \widehat{g}(S)\chi_S, \qquad T_\rho g = \sum_S \rho^{|S|}\widehat{g}(S)\chi_S.$$

Write $W_k(g) = \sum_{|S|=k} \widehat{g}(S)^2$. A balanced Boolean function satisfies

$$W_0 = 0, \quad \sum_{k \geq 1} W_k = 1, \quad \mathbb{E}[g T_\rho g] = \sum_{k \geq 1} \rho^k W_k, \quad \mathbb{E}(T_\rho g)^2 = \sum_{k \geq 1} \rho^{2k} W_k \leq \rho^2. \tag{7}$$

Let E_i average coordinate i, $D_i = I - E_i$, and $N = \sum_i D_i$. Then $N\chi_S = |S|\chi_S$ and $T_{e^{-t}} = e^{-tN}$. Since each D_i is a self-adjoint projection,

$$\mathbb{E} u N v = \sum_i \mathbb{E}[D_i u \, D_i v]. \tag{8}$$

On an edge with endpoint values $u_\pm, v_\pm$, its conditional contribution is $(u_+ - u_-)(v_+ - v_-)/4$. These conventions agree with the reference [10].

The pointwise sensitivity of a Boolean g is

$$S_g(x) = |\{i : g(x) \neq g(x^{(i)})\}|.$$

Since $D_i g = g \, 1_{\{g(x) \neq g(x^{(i)})\}}$,

$$Ng = g S_g, \qquad \mathbb{E} S_g = \sum_k k W_k, \qquad \mathbb{E} S_g^2 = \sum_k k^2 W_k. \tag{9}$$

2.2 Compression for concave objectives

Lemma 1 (Coordinate sorting)**.** *Let $K : [-1,1] \to \mathbb{R}$ be continuous and concave. For any Boolean g, there is an increasing Boolean $g^\uparrow$ of the same mean such that*

$$\mathbb{E} K(T_\rho g) \geq \mathbb{E} K(T_\rho g^\uparrow), \qquad 0 \leq \rho \leq 1.$$

The analogous statement holds for concave functions on $[0,1]$ and indicator inputs.

Proof. The proof is from [3, 4]. For one coordinate replace sections g_-, g_+ by their pointwise minimum and maximum. After noise on the other coordinates, the average of the two section outputs is unchanged. Their half-difference changes in absolute value from $|T_\rho(g_+ - g_-)|/2$ to $T_\rho|g_+ - g_-|/2$, which is at least as large by positivity of the noise kernel. For fixed average a, the function $z \mapsto K(a+z) + K(a-z)$ is even and concave, hence nonincreasing for $z \geq 0$. Sorting therefore does not increase the objective. It preserves monotonicity already obtained in other coordinates, because pointwise minima and maxima of increasing functions are increasing. Sort each coordinate once. □

3 A common entropy-kernel induction

For $1/2 \leq \alpha < 1$ put $\gamma = (1-\alpha)/\alpha$. We use

$$G(t) = F_\alpha(t)\left[1 + \frac{(2t-1)P(F_\alpha(t))}{P(F_\alpha(t)) + F_\alpha(t)^\gamma N(F_\alpha(t))}\right], \qquad P(0) = 1, \quad N(0) = d > 0, \tag{10}$$

where N, P are functions remaining to be determined. Positive N, P on the entropy range give $G > 0$ in the interior, $G(0) = G(1) = 0$, $G'(0+) = d$, and

$$G(t) + G(1-t) = 2F_\alpha(t). \tag{11}$$

The exponent γ gives the finite positive left slope because $F_\alpha(t) \sim t^\alpha$. For $\alpha > 1/2$, using F_α instead of F_α^γ in the correction would give $G(t) = o(t)$, incompatible with a nonzero nonnegative concave kernel. At $\alpha = 1/2$ we take $P = 1$, $N = \Lambda$. All choices below use the same induction; only the parameter row changes.

Fix $0 < p < 1/2$ and put $\rho = 1 - 2p$. Let $c, d > 0$ and let $G \geq 0$ be continuous on $[0,1]$ with $G(0) = G(1) = 0$. In the applications $d = G'(0+)$, but the criterion below only uses d as the positive normalization constant in the pair inequality. For fixed p, choose $0 \leq \phi \leq 1$, $\eta, \kappa > 0$ and a nonnegative polynomial B. Set $k = 2c/d$ and

$$b_*(m) = \min\{3/8, 9/16 - |m|/2, 1 - |m|\},$$
$$A = \frac{2B(m) - B(m+x) - B(m-x)}{2x^2}, \quad C = \frac{B(m+x) - B(m-x)}{2x},$$

using polynomial extensions at $x = 0$.

Proposition 1 (Common scalar-to-product criterion)**.** *Suppose the following four conditions hold. For every $0 \leq u \leq v \leq 1$, with $z = v - u$, let*

$$D = \frac{G(u+pz) + G(v-pz) - G(u) - G(v)}{2},$$
$$S = \phi G(v) + (1-\phi)G(u), \quad W = z - \frac{G(v) - G(u)}{d}.$$

The pair, closure, base and gate conditions are respectively

$$D \geq 0, \qquad D(S + \eta D) \geq \kappa c^2 W^2, \tag{12}$$
$$A[B(m) + (2\phi - 1)xC + (\eta - 1)x^2 A] \leq \kappa(1 - kC)^2 \quad (0 \leq x \leq b_*(m)), \tag{13}$$
$$\mathbb{E}G((1 + T_\rho g)/2) \geq cB(\mathbb{E}g) \quad \textit{for increasing three-bit } g, \tag{14}$$
$$1 - \frac{\max(\phi, 1-\phi)}{\eta} - k\sqrt{\frac{\kappa}{\eta}} > 0. \tag{15}$$

Then the base inequality in (14) *holds for every increasing Boolean sign g. If G is concave, it holds for every Boolean sign g.*

Here increasing three-bit g means that g has at most three active coordinates and meanwhile is increasing.

Proof. An increasing sign function of mean m with at least four active coordinates has an active coordinate of influence $0 < x \le b_*(m)$. To see this, choose four active coordinates $X_{i_1}, \dots, X_{i_4}$ and put $Z = (X_{i_1} + \cdots + X_{i_4})/4$. Then $\mathbb{E}[gZ]$ is the average of the four corresponding first Fourier coefficients, so the smallest positive first Fourier coefficient among all active coordinates is at most $\mathbb{E}[gZ]$. The variable Z takes the five values $-1, -1/2, 0, 1/2, 1$ with respective probabilities $(1, 4, 6, 4, 1)/16$. At fixed mean m, rearrangement maximizes $\mathbb{E}[gZ]$ by assigning $g = 1$ to the largest values of Z first. By symmetry of the distribution of Z, it suffices to consider $m < 0$. In this case, the resulting bound can be expressed using the minority mass $a = (1 - |m|)/2$ (at 1): filling the level $Z = 1$ gives $2a$; if $a > 1/16$, filling the level $Z = 1/2$ next gives $a + 1/16$; after all positive levels are filled, the bound is $3/8$. Thus $\mathbb{E}[gZ] \le \min\{2a, a + 1/16, 3/8\} = b_*(m)$. For a coordinate attaining $\mathbb{E}[gX_i] = x$, the two sections g_+, g_- have means $m + x$ and $m - x$, since $m = (\mathbb{E}g_+ + \mathbb{E}g_-)/2$ and $x = (\mathbb{E}g_+ - \mathbb{E}g_-)/2$.

Induct on the number of active coordinates, splitting along this coordinate. The noisy child probabilities $u = \frac{1+T_\rho g_-}{2}, v = \frac{1+T_\rho g_+}{2}$, satisfy $u \le v$ and $\mathbb{E}(v - u) = x$. Write

$$y = \frac{\mathbb{E}G(v)}{c}, \quad t = \frac{\mathbb{E}G(u)}{c}, \quad e = \frac{\mathbb{E}D}{c}, \quad Q = \phi y + (1 - \phi)t, \quad \xi = x - \frac{k}{2}(y - t).$$

Equation (12) gives $\sqrt{\kappa}\, c\, |W| \le \sqrt{D}\, \sqrt{S + \eta D}$. Taking expectation and applying Cauchy–Schwarz give $e(Q + \eta e) \ge \kappa \xi^2$. The decomposition gives

$$\frac{1}{c}\, \mathbb{E}\, G\left(\frac{1 + T_\rho g}{2}\right) = \frac{y + t}{2} + e.$$

Since $e \ge 0$, $Q \ge 0$, and $e(Q + \eta e) \ge \kappa \xi^2$, we have

$$e \ge \frac{\sqrt{Q^2 + 4\eta\kappa\xi^2} - Q}{2\eta}.$$

Substituting this lower bound for e yields

$$\frac{1}{c}\, \mathbb{E}\, G\left(\frac{1 + T_\rho g}{2}\right) \ge R_x(y, t) := \frac{y + t}{2} + \frac{\sqrt{Q^2 + 4\eta\kappa\xi^2} - Q}{2\eta}. \tag{16}$$

Each partial derivative of R_x is at least half the positive quantity in (15); continuity handles the boundary. Substitute the inductive bounds $y \ge B(m + x)$, $t \ge B(m - x)$. Their average is $B(m) - x^2 A$, while $Q = B(m) - x^2 A + (2\phi - 1)xC$ and $\xi = x(1 - kC)$. For $A \le 0$ the average suffices; for $A > 0$, (13) says exactly that the nonnegative root correction in (16) is at least $x^2 A$, since $Q \ge 0$. This closes the induction. If G is concave, apply the indicator-input version of lemma 1 to $f = (1 + g)/2$. It gives an increasing $f^\uparrow$ with $\mathbb{E}f^\uparrow = \mathbb{E}f$ and $\mathbb{E}G(T_\rho f) \ge \mathbb{E}G(T_\rho f^\uparrow)$. The inequality for increasing inputs therefore implies it for arbitrary Boolean g. □

Corollary 1 (Symmetric entropy comparison)**.** *Suppose the assumptions in proposition 1. In addition, suppose F is concave and symmetric on $[0, 1]$, $G(t) + G(1 - t) = 2F(t)$, $c = F(p)$ and $B(0) = 1$. Then every balanced Boolean f satisfies $\mathbb{E}F(T_\rho f) \ge F(p)$.*

Proof. For increasing balanced $g = 2f - 1$, apply proposition 1 to g and its increasing dual $-g(-\cdot)$, and average. The reflection identity gives the conclusion. Concavity of F and lemma 1 remove monotonicity. □

4 Proof of Theorem 1

For $\alpha = 1/2$ write $F = F_{1/2}$, $c = F(p) = \sqrt{1+s} - 1$ and $s = 2\sqrt{p(1-p)}$. Every kernel used here has the single form

$$G(t) = F(t)\left[1 + \frac{2t-1}{1 + F(t)\Lambda(F(t))}\right], \qquad d = \Lambda(0). \tag{17}$$

The four polynomial choices, named kernel families I–IV, are

$$\begin{aligned}
\Lambda_{\mathrm{I}}(x) &= (26 - 7x + 4x^2 + 31x^3 + 18x^4 + 3x^5)/20,\\
\Lambda_{\mathrm{II}}(x) &= (660 - 146x - 44x^2 + 529x^3 + 324x^4 + 54x^5)/500,\\
\Lambda_{\mathrm{III}}(x) &= (132612675 - 47653920x + 53479214x^2 + 102660617x^3)/10^8,\\
\Lambda_{\mathrm{IV}}(x) &= 1.32889710 - 1.02325943x + 3.03035249x^2 + 0.80706624x^3\\
&\qquad - 9.25961996x^4 - 8.97492640x^5 + 35.00400827x^6.
\end{aligned} \tag{18}$$

All decimals in this paper are exact. In all four cases $\Lambda > 0$ on $[0, \sqrt{2}-1]$; for IV, Bernstein coefficients on $[0, 5/12]$ give $\Lambda_{\mathrm{IV}} - 1 \geq 1916794877/8640000000 > 1/5$.

Lemma 2 (One certified interface for all square-root bands)**.** *Every complete row of Table 1 satisfies the hypotheses of Proposition 1, with*

$$d = \Lambda(0), \qquad k = 2c/d, \qquad \kappa = \mu s^2/c^2 = \mu(1+\sqrt{1+s})^2. \tag{19}$$

The profile B associated with each row is normalized by $B(0) = 1$ and specified explicitly in appendix B.6.

Proof. Appendix B gives a common pair normalization and a profile/base verification procedure. The certificate requirements for Families I–II are identified in appendix B.3; the final Family-II band uses the analytic propagation in appendix B.4, and the Family-III–IV records appear in appendix B.5. Substitution of (19) turns the certified pair and closure inequalities into those of Proposition 1. □

Proof of Theorem 1. Apply Corollary 1 to every row. Their union covers $\rho_0 = \sqrt{184}/25 \leq \rho \leq 1971/2000$: the Family I–II rows cover $.19 \leq s \leq .84$, the extra II row reaches $\rho_1 = \sqrt{150675679}/12500$, and

$$\rho_1^2 - (491/500)^2 = 27/78125000 > 0$$

gives the overlap with III. The remaining bands meet at their endpoints.

For $0 < \rho < \rho_0$ use the entropy-energy inequality of Lemma 5; see details in Appendix A. Put

$$\psi(u) = 4 - 2\sqrt{2}J(u), \qquad d_t = T_{e^{-t}}g, \qquad \psi(a(t)) = \mathbb{E}\psi(d_t), \qquad t_0 = -\log\rho_0. \tag{20}$$

The certified value at t_0 gives $a(t_0) \leq \rho_0$. Moreover,

$$\psi'(a)a' = -\sum_i \mathbb{E}[D_i d_t D_i \psi'(d_t)] \leq 0.$$

Hence $a(t) \leq \rho_0$ for $t \geq t_0$, so Lemma 5 applies throughout this trajectory and gives $a' \leq -a$ whenever $a > 0$. Consequently $a(t) \leq a(t_0)e^{-(t-t_0)} \leq e^{-t}$; the zero-amplitude case is immediate. Thus, for $t > t_0$,

$$\mathbb{E}\psi(T_\rho g) = \psi(a(t)) \leq \psi(e^{-t}) = \psi(\rho), \tag{21}$$

equivalently, for $\rho < \rho_0$,

$$\mathbb{E}J(T_\rho g) \geq J(\rho),$$

implying the conjecture for $0 < \rho < \rho_0$. The endpoints follow by continuity. In fact the proof establishes the slightly stronger bound $\rho \leq 1971/2000$, which will be used next. □

Table 1: Parameter rows. The last column is the highest Chebyshev index, not the degree of B.

Family	Parameter interval	ϕ	η	μ	d
I	$s \in [0.825, 0.84]$	0.505	1.2	0.209	16
I	$s \in [0.805, 0.825]$	0.51	1.2	0.209	16
I	$s \in [0.77, 0.805]$	0.54	1.2	0.209	16
I	$s \in [0.7, 0.77]$	0.54	1.2	0.209	16
I	$s \in [0.6, 0.7]$	0.58	1.18	0.209	14
I	$s \in [0.5, 0.6]$	0.64	1.16	0.2053	14
I	$s \in [0.49, 0.51]$	0.69	1.15	0.2015	12
I	$s \in [0.38, 0.49]$	0.65	1.1	0.203	18
I	$s \in [0.33, 0.38]$	0.65	1.1	0.203	18
I	$s \in [0.28, 0.33]$	0.65	1.1	0.203	18
II	$s \in [0.24, 0.28]$	0.7	1.15	0.2078	16
II	$s \in [0.21, 0.24]$	0.7	1.15	0.2078	18
II	$s \in [0.195, 0.21]$	0.7	1.15	0.2078	20
II	$s \in [0.19, 0.195]$	0.7	1.15	0.2075	22
II	$s \in [0.18888, 0.19]$	0.7	1.163806	0.20747737	22
III	$\rho \in [0.982, 0.984]$	0.723	1.066	0.2079	26
III	$\rho \in [0.984, 0.9846]$	0.723	1.066	0.2079	30
III	$\rho \in [0.9846, 0.9849]$	0.723	1.066	0.2079	32
III	$\rho \in [0.9849, 0.985]$	0.723	1.066	0.2079	32
IV	$\rho \in [0.985, 0.9855]$	0.725739	1.070345	0.20848	38

5 Proof of Theorem 2

We need the following recent result.

Theorem 3 (Hellinger contraction [6])**.** *For every Boolean $g : \{-1,1\}^n \to \{-1,1\}$ of mean m,*

$$H(m) - \mathbb{E}H(T_\rho g) \le 1 - H(\rho). \tag{22}$$

Thus $\mathbb{E}H(T_\rho g) \ge H(\rho)$ when g is balanced.

Put $\alpha_* = (5-\sqrt{13})/2$ and $\beta_a(y) = J_a(H^{-1}(y)) = J_a(\sqrt{1-y^2})$ for $y \in [0,1]$. For $1/2 < a < 1$, set $z = 1-a$ and

$$R_z(u) = \operatorname{arctanh} \frac{zu}{1-zu^2} - z \operatorname{arctanh} u.$$

The inverse-hyperbolic-tangent argument lies in $(0,1)$, and differentiation gives

$$\begin{aligned} \beta_a''(y) &= \frac{u(1-u^2)J_a''(u) - J_a'(u)}{u^3}, \qquad u = \sqrt{1-y^2}, \\ R_z'(u) &= \frac{zu^2[(z^2+3z-1) - z(1+z)u^2]}{(1-u^2)[(1-zu^2)^2 - z^2u^2]}, \end{aligned} \tag{23}$$

The sign of $\beta_a''(y)$ is opposite to that of $R_z(u)$: indeed its numerator is a positive multiple of $(1-zu^2)\tanh(z \operatorname{arctanh} u) - zu$. If $a \ge \alpha_*$, then $R_z' < 0$, so β_a is strictly convex. If $1/2 < a < \alpha_*$, then R_z' changes sign once from positive to negative; since $R_z(0) = 0$ and $R_z(u) \to -\infty$ at 1, β_a has exactly one inflection point c_a, being convex before it and concave after it. In both cases β_a is increasing on $[0,1]$.

The threshold is intrinsic to this convexity argument, since $\beta_a''(1) = 2^{1-a}a(a-1)(a^2-5a+3)/3 < 0$ for $a < \alpha_*$. At $a = 1/2$, $\beta_{1/2}(y) = \sqrt{1+y}$ is concave.

Lemma 3 (Upward transfer)**.** *For $0 < a < b < 1$, $0 \le r \le 1$, and any random variable $|U| \le 1$,*

$$\mathbb{E}J_a(U) \ge J_a(r) \quad \Longrightarrow \quad \mathbb{E}J_b(U) \ge J_b(r).$$

The map $\Gamma_{b,a} = J_b \circ J_a^{-1}$, with inverse on $[0,1]$, is increasing and strictly convex on $[1, 2^{1-a}]$.

Proof. For $u = \tanh x$ and $z = 1 - a$,

$$\frac{J_a''(u)}{J_a'(u)} = \frac{z\tanh x + z\coth(zx)}{1-u^2}.$$

Its numerator strictly increases in z, because $\tanh x + \coth(zx) - zx\,\mathrm{csch}^2(zx) > 0$. Thus $J_b''/J_b' < J_a''/J_a'$. Since both first derivatives are negative, the chain rule gives $\Gamma_{b,a}' > 0$ and $\Gamma_{b,a}'' > 0$. Jensen and monotonicity yield $\mathbb{E}J_b(U) \ge \Gamma_{b,a}(\mathbb{E}J_a(U)) \ge J_b(r)$. □

For $1/2 < a < \alpha_*$, define $\tau_a \in (0, c_a)$ by

$$\beta_a(1) - \beta_a(\tau_a) = (1-\tau_a)\beta_a'(\tau_a). \tag{24}$$

Existence and uniqueness follow from $M_a(y) = \beta_a(1) - \beta_a(y) - (1-y)\beta_a'(y)$: $M_a(0) > 0$, $M_a' = -(1-y)\beta_a''(y)$, and $M_a < 0$ on $[c_a, 1)$.

Lemma 4. *The greatest convex minorant β_a is*

$$\underline{\beta}_a(y) = \begin{cases} \beta_a(y), & y \le \tau_a, \\ \beta_a(\tau_a) + \beta_a'(\tau_a)(y - \tau_a), & y \ge \tau_a. \end{cases} \tag{25}$$

Consequently

$$\inf_{|U|\le 1,\ \mathbb{E}H(U)\ge s} \mathbb{E}J_a(U) = \underline{\beta}_a(s). \tag{26}$$

Take $a = 13/25$ in the common kernel (10), so $\gamma = 12/13$. Five pairs (N_i, P_i), with degrees $(6, 2)$, and five profiles $B_i = (1-m^2)Q_i$ are listed in Appendix C. Their crossover intervals are

$$\begin{aligned} I_1 &= [10^{-6}, 1/50000], & I_2 &= [1/50000, 1/2500], \\ I_3 &= [1/2500, 13/10000], & I_4 &= [13/10000, 7/2000], \\ I_5 &= [7/2000, 29/4000]. \end{aligned} \tag{27}$$

The common pair, closure, base, and gate conditions are certified on each complete band. All fifteen pair charts are complete, with 370716 accepted leaves; see appendix C.2. Thus Proposition 1 applies without a new induction argument or a change in normalization.

Proof of Theorem 2. First prove the assertion at $a = 13/25$. The square-root bound just proved and upward transfer give it for $\rho \le 1971/2000$. The induction (Corollary 1) with the five rows of (27) gives it for $1971/2000 \le \rho \le 499999/500000$; see details in Appendix C. For the remaining range set $p_0 = 10^{-6}$ and $s_0 = 2\sqrt{p_0(1-p_0)}$. The exact root certificate in appendix C.4 gives

$$\frac{832560563}{10^{12}} < \beta_a(1) - \beta_a(s_0) - (1-s_0)\beta_a'(s_0) < \frac{832560564}{10^{12}}. \tag{28}$$

The supporting-line criterion therefore gives $s_0 \le \tau_a$. For $U = T_\rho g$, $s = \sqrt{1-\rho^2}$, theorem 3 gives $\mathbb{E}H(U) \ge s$ and lemma 4 yields for $0 \le s \le s_0$,

$$\mathbb{E}J_a(U) \ge \underline{\beta}_a(s) = \beta_a(s) = J_a(\rho).$$

These three closed ranges cover $[0, 1]$. Upward transfer proves the claim for every $\alpha > a$, with equality classified by its strictness. □

Appendices

A An entropy-energy inequality for the square-root profile

The following entropy-energy inequality supplies the proof of Theorem 1 for the case $\rho \le \sqrt{184}/25$. Throughout this appendix, set

$$\begin{aligned}
\psi(x) &= 4 - 2\big(\sqrt{1+x} + \sqrt{1-x}\big), \qquad (29)\\
p(x) &= \psi'(x) = (1-x)^{-1/2} - (1+x)^{-1/2},\\
q(x) &= \psi''(x) = \frac{1}{2}\big((1-x)^{-3/2} + (1+x)^{-3/2}\big),\\
\mathcal{T}(x) &= \int_0^x \sqrt{q(t)}\, dt, \qquad -1 < x < 1. \qquad (30)
\end{aligned}$$

The function $\mathcal{T}$ is odd with finite endpoint limits. Define a variant of Dirichlet form:

$$\mathcal{E}_\psi(d) = \sum_i \mathbb{E}[D_i d\, D_i p(d)]. \qquad (31)$$

Lemma 5 (Entropy-energy inequality)**.** *Let $d : \{-1,1\}^n \to (-1,1)$ satisfy $\mathbb{E}d = 0$, and define $a \in [0,1)$ by $\mathbb{E}\psi(d) = \psi(a)$. If*

$$a^2 \le r_0 := \frac{184}{625} = 0.2944, \qquad (32)$$

then

$$\mathcal{E}_\psi(d) \ge a p(a). \qquad (33)$$

For $a > 0$, equality holds if and only if $d = \pm a x_i$ for a coordinate i. If $a = 0$, then $d = 0$.

The parameter a is fixed by the entropy level. At a crossing of the Boolean comparison it equals ρ; continuation therefore also needs the certified starting value from Section 4.

A.1 Properties of the auxiliary transform

This subsection proves Lemmas 6 and 7, which provide the convexity and derivative estimates used in Lemma 5. The polynomial calculation for Lemma 7 is included in its own proof.

For $0 \le v < \mathcal{T}(1)^2$, define

$$\Omega(v) = x p(x) - v, \qquad v = \mathcal{T}(x)^2, \quad x \ge 0. \qquad (34)$$

Lemma 6 (Convexity of the excess function)**.** *The function Ω is strictly convex on $(0, \mathcal{T}(1)^2)$, with $\Omega(0) = \Omega'(0) = 0$. In particular, it is nonnegative and increasing, and $\Omega(v)/v$ is increasing for $v > 0$.*

Proof. Differentiation yields

$$1 + \Omega'(\mathcal{T}(x)^2) = \frac{p(x) + x q(x)}{2\mathcal{T}(x)\sqrt{q(x)}}. \qquad (35)$$

Since $x q(x) \ge p(x)$, strict convexity is equivalent to

$$\mathcal{T}(x) > R(x) := \frac{2q(x)^{3/2}(p(x) + x q(x))}{4q(x)^2 + (x q(x) - p(x))q'(x)}. \qquad (36)$$

Put $x = 2t/(1+t^2)$, $z = t^2$, and

$$\begin{aligned}
A(z) &= 4(1-z)^2(1+3z)(2+z+z^2),\\
B(z) &= (1+z)\{4(1-z)^2(1+3z)^2 + 3z^2(5-z)(5+10z+z^2)\}.
\end{aligned}$$

Then $R = \sqrt{q}\, r$, where $r = tA(z)/B(z)$, and $\ell_0 = (\sqrt{q})'/\sqrt{q} = 3t(1+z)(5+10z+z^2)/[4(1-z)^2(1+3z)]$. Direct polynomial expansion gives

$$1 - \ell_0 r - \frac{dr}{dx} = \frac{12z^2 Q_\Omega(z)}{(1+3z)B(z)^2} > 0, \tag{37}$$

where

$$\begin{aligned} Q_\Omega(z) = {} & 100 + 920z + 3745z^2 + 9165z^3 + 14768z^4 + 16128z^5 \\ & + 12098z^6 + 6210z^7 + 2036z^8 + 336z^9 + 21z^{10} + 9z^{11}. \end{aligned}$$

Thus $\mathcal{T}' > R'$; both vanish at zero, proving the comparison. The expansions $\mathcal{T}(x) = x + O(x^3)$ and $p(x) = x + O(x^3)$ give $\Omega(0) = \Omega'(0) = 0$, and the other assertions follow by convexity. □

Lemma 7 (A derivative bound for the change of variables)**.** *For $v = \mathcal{T}(x)^2$ and $0 < x < 1$,*

$$0 \le \frac{d}{dv}\frac{\mathcal{T}(x)}{x} \le \frac{5}{16}. \tag{38}$$

The upper constant is attained in the limit $x \downarrow 0$.

Proof. For $b = \sqrt{q}$,

$$\frac{d}{d(\mathcal{T}^2)}\frac{\mathcal{T}}{x} = \frac{xb - \mathcal{T}}{2x^2 \mathcal{T} b} \ge 0.$$

The upper bound is equivalent to $\mathcal{T} \ge R_D := xb/[1 + (5/8)x^2 b]$. Differentiation shows that $\mathcal{T}' \ge R_D'$ follows from

$$\frac{b'}{b^2} \le \frac{15}{8}x + \frac{25}{64}x^3 b. \tag{39}$$

Put $y = \sqrt{1-x^2}$, $w = \sqrt{(1+y)/2}$, and

$$q = \frac{(2-y)w}{y^3}, \quad L = \frac{b'}{xb} = \frac{3(4+2y-y^2)}{4y^2(2-y)(1+y)}, \quad D = y^3(2-y)(1+y).$$

The required inequality is $N \le 120\sqrt{q}D$, where

$$N = 48y(4+2y-y^2) - 25w(1-y)(1+y)^2(2-y)^2. \tag{40}$$

If $w \le 4/5$, then $y \le 7/25$ and

$$\frac{48y(4+2y-y^2)}{25(1-y)(1+y)^2(2-y)^2} \le \frac{490175}{710016} < \frac{7}{10} < \frac{1}{\sqrt{2}} \le w,$$

so $N < 0$. The ratio is increasing because its numerator increases and its denominator has logarithmic derivative $-y(7-5y)/[(1-y)(1+y)(2-y)] \le 0$. If $w \ge 4/5$, substitute $y = 2w^2 - 1$ and define

$$P = 14400wy^3(2-y)^3(1+y)^2 - N^2, \quad Q_D = P/[64(1-w)], \quad C_D(t) = 5^{17}Q_D((4+t)/5).$$

Exact expansion gives a polynomial C_D of degree 21, whose only negative coefficients occur at degrees $5, 6, 7$. Its coefficients at $4, 5, 6, 7$ are

$$515636013679740, \quad -129753241411356, \quad -202277501092524, \quad -64518339810120.$$

Their sum is $119086931365740 > 0$; hence $C_D \ge 0$ on $[0,1]$ using $t^5, t^6, t^7 \le t^4$. Thus $N^2 \le 14400qD^2$. The exact expansion is checked by our codes. Integrating $\mathcal{T}' \ge R_D'$ from zero proves the bound; $\mathcal{T}(x) = x + (5/16)x^3 + O(x^5)$ gives the limiting constant. □

A.2 A strong scalar remainder inequality

This subsection proves Lemma 8 (the bounds for h) and Lemma 9 (the strong remainder inequality). Their polynomial positivity certificates are recorded separately in Subsection A.5.

Introduce, for $0 \le u \le 1$,

$$A_0(u) = (2-u^2)\sqrt{\frac{1+u^2}{2}}, \qquad h(u) = \sqrt{A_0(u)}\,\frac{\mathcal{T}(\sqrt{1-u^4})}{\sqrt{1-u^4}}. \tag{41}$$

Endpoint values are understood by continuity.

Lemma 8 (Bounds for the comparison function)**.** *For $z = 1-u \in [0,1]$,*

$$1 + 2z + \frac{3}{8}z^2 \le h(u) \le 1 + 2z + \frac{57}{100}z^2, \qquad -h'(u) \ge 2 + \frac{13}{20}z. \tag{42}$$

Proof. Differentiation gives

$$(1-u^4)h' = \frac{u^3(5-u^2)}{2(2-u^2)}h - 2A_0, \qquad h(1) = 1. \tag{43}$$

Set

$$\begin{aligned} P_c &= 1 + 2(1-u) + c(1-u)^2, & D_0 &= 2(2-u^2),\\ C_0 &= u^3(5-u^2), & B_0 &= 8(2-u^2)^4(1+u^2),\\ R_c &= D_0(1-u^4)[2 + 2c(1-u)] + C_0P_c,\\ R_* &= D_0(1-u^4)[2 + (13/20)(1-u)] + C_0P_{57/100}. \end{aligned}$$

The certificates in appendix A.5 give $B_0 \ge R_{3/8}^2$, $R_{57/100}^2 \ge B_0$, and $B_0 \ge R_*^2$. Since all R's are positive, backward comparison in (43) gives $P_{3/8} \le h \le P_{57/100}$, and substitution of the upper bound gives $-h' \ge 2 + (13/20)(1-u)$. At the singular endpoint the integrating factor is of order $\sqrt{1-u}$, so its product with the bounded comparison difference tends to zero; the same comparison remains valid. □

Lemma 9 (Strong remainder inequality)**.** *Assume $a^2 \le r_0$, set $M = \mathcal{T}(a)^2$, and put*

$$\alpha = \frac{2\mathcal{T}(a)\sqrt{q(a)}}{p(a)}$$

for $a > 0$, with $\alpha = 2$ at $a = 0$. For every $x \in [-1,1]$,

$$\mathcal{T}(x)^2 - M - \alpha\big(\psi(x) - \psi(a)\big) \ge \frac{153}{2000}\big(\mathcal{T}(x)^2 - M\big)^2. \tag{44}$$

The inequality is strict unless $|x| = a$.

Proof. Set $u = (1-x^2)^{1/4}$, $y = (1-a^2)^{1/4} \ge \sqrt{.84}$, and $c(t) = t/\sqrt{1+t^2}$. Then $(\mathcal{T}(\sqrt{1-u^4})^2)' = -4h(u)$, $(\psi(\sqrt{1-u^4}))' = -2\sqrt{2}c(u)$, and $\alpha = \sqrt{2}h(y)/c(y)$. Write $\mathcal{D}_a(x)$ for the left side of the claimed inequality. Then

$$\mathcal{D}_a(x) = 4\int_u^y [h(t) - h(y)]\,dt + \frac{4h(y)}{c(y)}\int_u^y [c(y) - c(t)]\,dt. \tag{45}$$

The scalar certificate below gives

$$c'(t) = (1+t^2)^{-3/2} \ge \frac{5-3t}{4\sqrt{2}}. \tag{46}$$

Integrate this and lemma 8 twice, for either order of u, y:

$$\mathcal{D}_a(x) \geq (u-y)^2 \left[4 + \frac{13}{30}(3-2y-u) + \frac{h(y)}{\sqrt{2}c(y)} \left(\frac{5}{2} - y - \frac{u}{2} \right) \right]. \tag{47}$$

Since $\sqrt{1+y^2}/(\sqrt{2}y) \geq (3-y)/2$, let $b = 1-u \in [0,1]$ and $e = 1-y \in [0, 17/200]$. The bracket is at least

$$R_B = 4 + \frac{13b}{30} + \frac{13e}{15} + \left(1 + \frac{5}{2}e + \frac{11}{8}e^2 + \frac{3}{16}e^3 \right) \left(1 + \frac{b}{2} + e \right). \tag{48}$$

The upper bound for h similarly gives

$$|\mathcal{T}(x)^2 - M| \leq 4|u-y| B_B, \qquad B_B = 1 + b + e + \frac{19}{100}(b^2 + be + e^2). \tag{49}$$

The exact positive certificate for

$$R_B - 16\frac{153}{2000} B_B^2 \tag{50}$$

on this rectangle proves the assertion, strictly for $u \neq y$. □

A.3 Variance gain and quantitative bounds

Here Lemma 10 derives the variance gain from Lemma 9 and the derivative estimate, while Lemma 11 supplies the quantitative bounds needed in the proof of Lemma 5. The polynomial check for Lemma 11 appears in Subsection A.5.

Lemma 10 (Variance gain)**.** *Under the hypotheses of Lemma 5, let*

$$g = \mathcal{T}(d), \qquad M = \mathcal{T}(a)^2, \qquad V = \mathbb{E}(g^2 - M)^2.$$

Then

$$\operatorname{Var} g - M \geq CV, \qquad C = \frac{191}{4000} = .04775. \tag{51}$$

It is strict unless $|d| = a$ everywhere. In addition,

$$|\mathbb{E}g|^2 \leq \frac{25}{256} a^2 V. \tag{52}$$

Proof. Positive even Taylor coefficients of ψ imply convexity of $v \mapsto \psi(\sqrt{v})$, hence $\mathbb{E}d^2 \leq a^2$. Averaging (44) gives $\mathbb{E}g^2 - M \geq (153/2000)V$. Since $\mathbb{E}d = 0$, lemma 7 gives

$$|\mathbb{E}g| = \left| \mathbb{E}\left[g - \frac{\mathcal{T}(a)}{a} d \right] \right| \leq \frac{5}{16} \mathbb{E}[|d|\,|g^2 - M|] \leq \frac{5}{16} a\sqrt{V}.$$

Subtract its square and use $153/2000 - (25/256)(184/625) = 191/4000$. Strictness follows from lemma 9; $\mathcal{T}(a)/a$ is interpreted continuously at zero. □

Lemma 11 (Bounds for the excess function)**.** *For $0 < a^2 \leq r_0$, let $M = \mathcal{T}(a)^2$ and $P = \Omega'(M)$. Then*

$$\frac{P}{M} \leq \frac{a^2}{4}, \qquad P < \frac{27}{1000}, \qquad \frac{\Omega(M)}{M} < \epsilon := \frac{19}{2000}. \tag{53}$$

Proof. First, for $0 \leq x < 1$,

$$\mathcal{T}(x) \geq x Q_F(x^2), \quad Q_F(r) = 1 + \frac{5}{16}r + \frac{81}{512}r^2 + \frac{5937}{57344}r^3 + \frac{39435}{524288}r^4. \tag{54}$$

Indeed the square of the derivative matches q through degree 8, and the differences at degrees 10, 12, 14, 16 are

$$\frac{5386809}{4194304},\quad \frac{326594687}{134217728},\quad \frac{7199920845}{2147483648},\quad \frac{1143517126455}{274877906944}.$$

All later coefficients of q are positive; integration proves the bound. For $r = a^2$, $s = \sqrt{1-r}$, $w = \sqrt{(1+s)/2}$, $p(a)/a = 1/(sw)$ and $q(a) = w(2-s)/s^3$. By (35), the first assertion follows from

$$\frac{s^2+s+2}{4Q_F(r)w^{3/2}s^{3/2}\sqrt{2-s}} \le 1 + \frac{1}{4}r^2 Q_F(r)^2. \tag{55}$$

Squaring twice, with positive sides, reduces this to

$$32Q_F(r)^4 s^6 (2-s)^2(1+s)^3\left(1+\frac{1}{4}r^2Q_F(r)^2\right)^4 - (s^2+s+2)^4 \ge 0, \quad r = 1-s^2, \tag{56}$$

certified below for $21/25 \le s \le 1$. It gives $P \le r^2 Q_F(r)^2/4 \le rM/4$. At $a_0^2 = 184/625$, lemma 8 and the rational lower bounds $\sqrt{.84} > .9165$, $\sqrt{.92} > .9591$ give

$$\mathcal{T}(a_0)^2 \le \frac{(184/625)[1 + 2(167/2000) + (57/100)(167/2000)^2]^2}{(29/25)(9591/10000)} < .363. \tag{57}$$

Thus $P < .027$. Finally $\Omega(M)/M$ increases with M, and $\sqrt{.92} > 23979/25000$ gives

$$\frac{\Omega(\mathcal{T}(a_0)^2)}{\mathcal{T}(a_0)^2} \le \frac{1}{(21/25)(23979/25000)Q_F(184/625)^2} - 1 < .0095.$$

All terminal comparisons are rational. □

A.4 Centered edges and Fourier concentration

This subsection proves the edge comparison in Lemma 12 and then completes the proof of the entropy-energy inequality, Lemma 5, used for the low-correlation part of Theorem 1.

Lemma 12 (Centered-edge comparison)**.** *For every coordinate edge,*

$$D_i d\, D_i p(d) \ge (D_i\mathcal{T}(d))^2 + \Omega\big((D_i\mathcal{T}(d))^2\big). \tag{58}$$

Proof. For $v = \mathcal{T}(x)$ put $\ell(v) = \log\sqrt{q(x)}$. It is even and

$$\ell''(v) = \frac{2qq'' - 3(q')^2}{4q^3} > 0.$$

To see the sign, weight the logarithmic slopes $s_\pm = \pm 3/[2(1 \mp x)]$ by the two summands of q. Then $q'/q = \mathbb{E}s$, $q''/q = (5/3)\mathbb{E}s^2$, giving positivity. For transformed edge endpoints spanning I, the energy divided by $(D_i\mathcal{T}(d))^2$ is

$$\left(\frac{1}{|I|}\int_I e^{\ell(v)}\,dv\right)\left(\frac{1}{|I|}\int_I e^{-\ell(v)}\,dv\right) = \mathbb{E}\cosh(\ell(U) - \ell(V)),$$

where U, V are independent uniform points of I. Conditionally on $|U - V| = \delta$, their midpoint is uniform on an interval centered at the center of I. Since $\ell(c + \delta/2) - \ell(c - \delta/2)$ is odd and increasing, its hyperbolic cosine is even and increasing in $|c|$. Centering I therefore minimizes the ratio. For endpoints $\pm\mathcal{T}(x)$ the energy is $xp(x)$ and the squared half-difference is $\mathcal{T}(x)^2$, proving the claim. Zero differences follow by continuity. □

Proof of Lemma 5. The case $a = 0$ is immediate. For $a > 0$ put $g = \mathcal{T}(d)$, $M = \mathcal{T}(a)^2$, $V = \mathbb{E}(g^2 - M)^2$, and suppose

$$\mathcal{E}_\psi(d) \le K := ap(a) = M + \Omega(M). \tag{59}$$

Write $q_1 = \sum_i \widehat{g}(i)^2$, $\ell = \max_i \widehat{g}(i)^2$, $H_0 = \operatorname{Var} g - q_1$, and $\mu = \mathbb{E}g$. Counting Fourier degrees and applying Jensen to the coordinate attaining ℓ, lemma 12 gives

$$\mathcal{E}_\psi(d) \ge \operatorname{Var} g + H_0 + \Omega(\ell). \tag{60}$$

Together with the variance gain this yields

$$H_0 + CV \le \Omega(M), \qquad q_1 \ge M - \Omega(M) + 2CV \ge (1 - \epsilon)M > 0, \tag{61}$$

and (52) implies

$$\frac{\mu^2}{q_1} \le \frac{(25/256)(184/625)\epsilon}{C(1-\epsilon)} < .077^2. \tag{62}$$

To bound the spread of the first level, let $L = \sum_i \widehat{g}(i)x_i$, $P_2 = L^2 - q_1$, and $W = \mathbb{E}P_2^2 = 2(q_1^2 - \sum_i \widehat{g}(i)^4)$. Then

$$q_1 - \ell \le \frac{W}{2q_1}. \tag{63}$$

Decompose $g = \mu + L + Z$, where $\mathbb{E}Z^2 = H_0$ and Z has degree at least 2. The degree-three part of LP_2 has squared norm $36\sum_{i<j<k}\widehat{g}(i)^2\widehat{g}(j)^2\widehat{g}(k)^2 \le 3q_1W$. Expanding $\mathbb{E}[g^2P_2]$, using $P_2 \ge -q_1$ and Cauchy–Schwarz, gives

$$\sqrt{VW} \ge W - (2\sqrt{3q_1} + 2|\mu|)\sqrt{WH_0} - q_1H_0. \tag{64}$$

Its positive root, with (62), satisfies

$$\sqrt{W} \le \sqrt{V} + 3.88\sqrt{q_1H_0}, \tag{65}$$

since $\sqrt{3} < 1.73206$ and $3.88^2 - (2(1.73206) + .154)3.88 - 1 = 10059/625000 > 0$. Now set $P = \Omega'(M)$. The tangent inequality strengthens (60) to

$$\mathcal{E}_\psi(d) \ge K + (1+P)(\operatorname{Var} g - M) + (1-P)H_0 - P(q_1 - \ell). \tag{66}$$

Lemma 11 imply

$$\begin{aligned}\frac{P}{(1+P)q_1C} + \frac{P(3.88)^2}{1-P} &\le \frac{184/625}{4(1-\epsilon)C} + \frac{.027(3.88)^2}{1-.027} \\ &= \frac{64883786279}{32870980625} < 2.\end{aligned} \tag{67}$$

Weighted Cauchy–Schwarz in (65), then (63), gives $P(q_1 - \ell) \le (1+P)CV + (1-P)H_0$. Substitution in (66) contradicts the assumed upper bound whenever $|d| \not\equiv a$, by the strict variance gain. If $|d| \equiv a$, the cube Poincaré inequality gives $\mathcal{E}_\psi(d) = (p(a)/a)\sum_i \mathbb{E}(D_i d)^2 \ge ap(a)$. Equality forces degree one, and the constant identity $d^2 = a^2$ forces one nonzero linear coefficient, hence $d = \pm ax_i$; these attain equality. □

A.5 Polynomial positivity certificates for Lemmas 8, 9, and 11

This subsection records precisely the finite polynomial positivity checks used in the proofs of Lemmas 8, 9, and 11. The separate polynomial check for Lemma 7 is given in Subsection A.1, not here.

For a polynomial $P(t) = \sum_{k=0}^N c_k t^k$, its Bernstein coefficients of order N are

$$\beta_i = \sum_{k=0}^{i} c_k \frac{\binom{i}{k}}{\binom{N}{k}}. \tag{68}$$

The polynomial is bounded below by $\min_i \beta_i$ on $[0,1]$, since Bernstein basis functions are nonnegative and sum to one. For other intervals we first make an affine substitution. The bivariate analogue uses the product of the two binomial ratios. Thus all checks below are finite rational computations.

Certificate for Lemma 8. The first four Bernstein checks establish $B_0 \geq R_{3/8}^2$, $R_{57/100}^2 \geq B_0$, and $B_0 \geq R_*^2$, which are used in the proof of (42). Use D_0, C_0, R_c, B_0, R_* defined in that lemma's proof, substitute $u = 1 - z$, and divide out the indicated power of z. The resulting certificates are:

Polynomial after substitution	Degree	z interval	$\lfloor 10^6 \min \beta \rfloor$
$(B_0 - R_{3/8}^2)/z^3$	11	$[0, 1]$	7000000
$(R_{57/100}^2 - B_0)/z^2$	12	$[0, 1/2]$	1585784
$(R_{57/100}^2 - B_0)/z^2$	12	$[1/2, 1]$	1447469
$(B_0 - R_*^2)/z^2$	12	$[0, 1]$	160000

Certificates for Lemma 9. The next two checks support the strong remainder inequality (44): first the derivative estimate (46), and then positivity of (50) on its full parameter rectangle. For (46), the degree-six polynomial

$$\frac{32 - (5 - 3t)^2(1 + t^2)^3}{(1 - t)^2}$$

has, in the variable $1 - t$, the Bernstein coefficients

$$72, \quad \frac{172}{3}, \quad \frac{628}{15}, \quad \frac{152}{5}, \quad \frac{322}{15}, \quad \frac{43}{3}, \quad 7.$$

For the bivariate polynomial (50), substitute $e = (17/200)t$. Its Bernstein coefficients of bidegree $(4, 4)$ have row minima satisfying

$$\left\lfloor 10^6 \min_j \beta_{ij} \right\rfloor = (3776000, 3397333, 2737146, 1679160, 11812).$$

Thus its sign is strict on the entire closed rectangle.

Certificate for Lemma 11. The final check proves the polynomial inequality (56), used to establish (55) and hence the first bound in (53). Substitute $s = 1 - z$ and $r = 2z - z^2$ in (56). The resulting degree-123 polynomial is divisible by z^2. After division and substitution $z = (4/25)t$, all degree-121 Bernstein coefficients are strictly positive, with

$$\left\lfloor 10^6 \min_i \beta_i \right\rfloor = 10182120.$$

This establishes (55) on its full parameter interval, including the removable equality endpoint $s = 1$.

The code source archive records exact-arithmetic checks, the associated convexity, and derivative verifiers. They reconstruct these polynomials and test their Bernstein coefficients and rational bounds. Their completed records remain inherited proof inputs.

B Certificates for Theorem 1

This appendix supports Lemma 2 by verifying the four conditions of Proposition 1 for the kernels in (17)–(18). Subsection B.1 normalizes the pair inequality and establishes the kernel concavity checks; B.2 describes the exact verification rules; B.3 records the requirements for the original Families I and II; B.4 propagates the final Family-II pair certificate analytically; B.5 gives the Family-III and IV certificates; and B.6 specifies all twenty exact mean profiles. Use $d_0 = \Lambda(0)$ and $L = d_0/2$ in the pair normalization and its verification.

B.1 Normalization of the Pair Inequality

The left, right, and cross charts below reduce the pair condition (12) of Proposition 1 to nondegenerate positivity tests. Their diagonal limits also establish the concavity of the corresponding kernels, used to infer $D \geq 0$.

Write $f[a, b]$ and $f[a, b, c]$ for continuous divided differences. For $z = v - u$, put

$$K = -\frac{G[u, u + pz, v] + G[u, v - pz, v]}{8}, \qquad W_0 = 1 - \frac{G[u, v]}{d_0}.$$

Exact interpolation gives $D_p = s^2 z^2 K$ and $W = zW_0$; hence the pair assertion is equivalent, away from degenerate pairs, to

$$\Gamma = K(S + \eta D_p) - \mu W_0^2 \geq 0. \tag{69}$$

Set $q_y = \sqrt{1 - y^2}$, $b_y = 1 - y/(1 + q_y)$, $f_y = yb_y = F(y^2)$, $l_y = \Lambda(f_y)$, $D_y = 1 + f_y l_y$, and $\Lambda_1(x) = (\Lambda(x) - d_0)/x$, extended polynomially at zero. The cancellation-free functions

$$\begin{aligned} N(y) &= \frac{b_y(2y + b_y l_y)}{D_y}, \qquad R(y) = \frac{b_y(2q_y^2 + f_y l_y)}{D_y}, \\ M(y) &= \frac{d_0(1 + b_y)/(1 + q_y) - b_y^3 \Lambda_1(f_y) + d_0 b_y l_y - 2b_y}{D_y} \end{aligned} \tag{70}$$

satisfy, by multiplication,

$$G(y^2) = y^2 N(y) = d_0 y^2 - y^3 M(y), \qquad G(1 - y^2) = yR(y). \tag{71}$$

They apply to every polynomial Λ above. The legacy checkers use algebraically identical expressions in the coordinate $f_y(f_y + 2)$.

Endpoint charts. Use $(u, v) = (a^2 b^2, b^2)$ on the left and $(u, v) = (1 - b^2, 1 - a^2 b^2)$ on the right, with $0 \leq a \leq 1, 0 \leq b \leq 4/5$. Write

$$d = j(2a + (1 - a)j), \quad w = a + (1 - a)j, \quad t = \sqrt{1 - (1 - a)d}, \quad p = d/(1 + a).$$

For a required interval $[p_-, p_+]$, $0 \leq j \leq \sqrt{p_+}$ covers all interpolants; monotone corner bounds exclude boxes missing that interval. Let m^i be the divided-difference table of $r \mapsto M(br)$ on the left or $r \mapsto R(br)$ on the right, at $(a, i, 1)$ for $i = w, t$. Its entries are $0, 1, 2, 01, 02, 12, 012$, with $0, 2, 02$ common. Put $E(a, b) = a^2/(a + b)$ and $E(0, 0) = 0$. On the left define

$$\begin{aligned} T &= (1 + a^2/(1 + a))m_2 + a^3 m_{02}/(1 + a), \\ S_0 &= \phi(d_0 - bm_2) + (1 - \phi)a^2(d_0 - bam_0), \\ P_i &= \frac{(1 - E(a, i)/(1 + a))m_2 + (i + E(a, i))m_{12}^i + E(a, i)a(m_{012}^i - m_{02}/(1 + a))}{1 + i}, \\ Z &= (P_w + P_t)/8. \end{aligned}$$

The left normalized certificate is

$$ZS_0 + 4\eta d(1 - a)(1 - w^2)bZ^2 - \mu bT^2/d_0^2 > 0. \tag{72}$$

On the right define

$$\begin{aligned} T &= (m_2 + am_{02})/(1 + a), \\ Z &= \frac{1}{8}\left[\frac{T - m_{12}^w - am_{012}^w}{1 + w} + \frac{a + w}{a + t}\frac{T - m_{12}^t - am_{012}^t}{1 + t}\right], \\ S_1 &= (\phi - \eta/2)am_0 + (1 - \phi - \eta/2)m_2 + (\eta/2)(wm_1^w + tm_1^t). \end{aligned}$$

The right normalized certificate is

$$ZS_1 - \mu(a+w)(b+T/d_0)^2 > 0. \tag{73}$$

The product rule and (71) identify these expressions with Γ/b and $b^2(a+w)\Gamma$, respectively. At $a = 1$ they give $K = -G''/8 > 0$, because $G > 0$. The two diagonal ranges $(0, .64]$ and $[.36, 1)$ cover $(0, 1)$; thus strict concavity is established before it is used below.

Cross chart. Every remaining pair has $u = l^2$, $v = 1 - a^2$ with $0 \le l, a \le 3/5$. Put

$$z = 1 - l^2 - a^2 \ge .28, \quad b = pz, \quad l_+ = \sqrt{l^2 + b}, \quad a_+ = \sqrt{a^2 + b},$$

$$P = N(l_+) + E(l, l_+)N[l, l_+], \qquad Q = R(a_+) + aR[a, a_+],$$
$$Z = \frac{Q + (a + a_+)P}{8(1-p)z}, \qquad W_0 = 1 - \frac{aR(a) - l^2N(l)}{d_0 z}.$$

Here $Z = (a + a_+)K > 0$. Certify $Y_0 > 0$ or $(Y_1 > 0, Y_2 > 0)$, where

$$Y_0 = Z[(\phi - \eta/2)aR(a) + (\eta/2)a_+R(a_+) + (1-\phi)l^2N(l) + (\eta/2)bP] - \mu(a + a_+)W_0^2,$$
$$Y_1 = Z[(\phi/2)R(a) + (\eta/2)(a_+ - a)P] - \mu W_0^2,$$
$$Y_2 = Z[(\eta/2)Q + (\eta/2)(a_+ - a)P] - \mu W_0^2.$$

The first is $(a+a_+)\Gamma$. For the alternative, combine Y_1, Y_2 with weights $2a/(a+a_+)$ and $(a_+ - a)/(a+a_+)$; the omitted part of the target bracket is $(1-\phi)l^2N(l)/(a + a_+) \ge 0$. Degenerate pairs follow by continuity.

B.2 Exact Verification Procedures

These rules validate all four hypotheses of Proposition 1, as required by Lemma 2: the pair checks from B.1, closure (13), the three-bit base cases (14), and the monotonicity gate (15). The algebraic endpoint identities are not separate numerical positivity tests.

Divided-difference tables propagate through sums, products, reciprocals and roots by exact identities. In particular, for $r = 1/f$ and $h = \sqrt{f}$,

$$r_{ij} = -f_{ij}r_ir_j, \quad r_{012} = -r_0(f_{012}r_2 + f_{01}r_{12}), \qquad h_{ij} = \frac{f_{ij}}{h_i + h_j}, \quad h_{012} = \frac{f_{012} - h_{01}h_{12}}{h_0 + h_2}.$$

For $E(a, b)$, $dE = h(2-h)\,da - h^2\,db$ with $h = a/(a+b) \in [0, 1/2]$; this also gives the removable-corner Lipschitz bounds. A box passes by outward interval positivity or by a rigorous center lower bound minus full-box derivative bounds times both actual endpoint radii. A domain or magnitude failure triggers subdivision, never acceptance. Each leaf must be positive or rigorously excluded; unresolved leaves fail. The legacy pair checkers use scale 10^{12} and guarded signed 128-bit integer arithmetic. The cubic and degree-six checkers use scale 2^{80} and signed 256-bit integers, with interval endpoints bounded by 2^{120}. All roots are bounded by exact square comparisons.

For profile verification, put $w = \sqrt{1+s}$ and $L = d_0/2$. The closure term has the common polynomial form

$$\mu\left[1 + w - \frac{w^2 - 1}{L}C\right]^2. \tag{74}$$

The legacy rows use degree-four Bernstein coefficients in w. For the later rows, the certified bound $B' \le 3$ implies $C \le 3$. The bracket is positive; as a function of w it is concave if $C \ge 0$ and increasing

if $C < 0$. Its squared minimum therefore occurs at one of the two rationally enclosed endpoints. In either method the resulting polynomial signs are verified on the same six squares:

$$m = (m_0 + d_m u)/16, \quad x = v(b_0 + d_b u)/16, \quad (u, v) \in [0, 1]^2,$$

$$(m_0, d_m, b_0, d_b) = (-16, 2, 0, 2), (-14, 8, 2, 4), (-6, 6, 6, 0), (0, 6, 6, 0), (6, 8, 6, -4), (14, 2, 2, -2).$$

For a polynomial $\sum a_{ij} u^i v^j$ of bidegree (n, m) its Bernstein coefficients are

$$b_{kl} = \sum_{i \le k, j \le l} a_{ij} \frac{\binom{k}{i}}{\binom{n}{i}} \frac{\binom{l}{j}}{\binom{m}{j}}.$$

After clearing positive denominators, these signs and midpoint de Casteljau subdivisions use integers only. Univariate versions certify $B/(1 - m^2) > 0$ and any derivative bounds. The endpoints and normalization are polynomial identities; the monotonicity gate is checked by squaring positive quantities. The base checker retains exactly the 20 increasing three-bit tables among all 256 truth tables. Constants and the three dictators are symbolic equalities; the other 15 tables are verified by outward intervals over the complete noise band, with exact root bounds.

B.3 Certificates for Families I and II

This subsection identifies the checks supporting Lemma 2 for the first fourteen rows of Table 1. The fifteenth (last Family-II) row is treated separately in appendix B.4. The kernel formulas and the individual mean profiles are given in (18) and appendix B.6.

Pair inequality and kernel concavity. The pair condition (12) is checked with the left, right, and cross charts in B.1, using the normalization (69). Rows sharing the same kernel and (ϕ, η, μ) can be grouped into the following nine intervals; each group requires positivity throughout its full s-interval, not merely at the endpoints:

Family	s interval	ϕ	η	μ
I	$[0.825, 0.84]$	0.505	1.2	0.209
I	$[0.805, 0.825]$	0.51	1.2	0.209
I	$[0.7, 0.805]$	0.54	1.2	0.209
I	$[0.6, 0.7]$	0.58	1.18	0.209
I	$[0.5, 0.6]$	0.64	1.16	0.2053
I	$[0.49, 0.51]$	0.69	1.15	0.2015
I	$[0.28, 0.49]$	0.65	1.1	0.203
II	$[0.195, 0.28]$	0.7	1.15	0.2078
II	$[0.19, 0.195]$	0.7	1.15	0.2075

The inherited verification record reports 21 completed Family-I and 6 completed Family-II pair-region checks, matching the 7×3 and 2×3 charts required by this grouping. The diagonal limits of the normalized inequalities also provide the kernel-concavity checks described in B.1. The region counts summarize reported computations; independent verification requires matching the original checker inputs and completion records to the nine bands above.

Profile positivity, closure, base, and gate. Unlike the pair groups, each of the first fourteen rows has its own profile B in appendix B.6. For every row, the nontrivial conditions to check are the profile positivity, closure (13) over the entire (m, x)-domain, and base (14) for the fifteen nontrivial increasing three-bit functions throughout the stated noise band. The two constants and three dictators are symbolic

base equalities. The normalization $B(0) = 1$ and the endpoint values $B(\pm 1) = 0$ are identities, not numerical certificates. The gate (15) is checked using the exact parameters in Table 1. B.2 specifies the Bernstein and outward-interval acceptance rules for these checks; Appendix D describes their reproduction. The present manuscript does not tabulate individual legacy row-by-row margins or run identifiers, so the general rules and the group counts alone should not be mistaken for independent completion records.

B.4 Analytic Extension of the Family II Certificates

This subsection justifies the pair condition (12) for the final Family-II row in Table 1, using a previously certified band. It is the analytic continuation step invoked in Lemma 2; the closure and base checks for the new row are covered by the common procedure in B.2.

Let $s_0 = 19/100$, $s_1 = 2361/12500$, $p_i = (1 - \sqrt{1 - s_i^2})/2$, and $q_i = 1 - p_i$. This row retains the Family-II kernel and $\phi = 7/10$, but uses

$$\eta = 581903/500000, \qquad \mu = 20747737/10^8.$$

No new pair search is needed: concavity in p gives $D_p \geq (p/p_0)D_{p_0}$ for $p_1 \leq p \leq p_0$. Thus the old constants $(\eta_0, \mu_0) = (23/20, 83/400)$ imply the new pair bound whenever

$$\eta \geq \eta_0 p_0/p_1, \qquad \mu \leq \mu_0 q_0/q_1, \tag{75}$$

since

$$D_p(S + \eta D_p) \geq (p/p_0)D_{p_0}(S + \eta_0 D_{p_0}) \geq \mu_0(q_0/(1-p))s^2W^2.$$

Both strict parameter margins and the gate have exact positive rational certificates. The endpoint is $\rho_1 = \sqrt{150675679}/12500$; the overlap is exact because $\rho_1^2 - (491/500)^2 = 27/78125000 > 0$.

B.5 Certificates for Families III and IV

The following records support Lemma 2 on the Family-III and IV bands: the three pair charts certify (12), and the profile/base calculations certify (13) and (14). The gate is checked by the procedure in B.2.

For kernels III and IV, use the parameters of Table 1. The common pair checks have the following completed records:

	positive leaves			excluded leaves		
kernel	left	right	cross	left	right	cross
cubic	2374	4686	2011	10	646	0
degree six	5466	5027	1558	57	395	0

Maximum depths are respectively $(20, 24, 21)$ and $(22, 24, 19)$. The legacy Family-I–II checks are described separately in appendix B.3. No recorded run for the Family-III–IV pair charts has unresolved leaves.

The common profile/base verification yields the following downward rounded lower bounds for the original unscaled expressions:

Row	closure leaves	closure gap	base leaves	base gap
Propagated	114	$1.548 \cdot 10^{-6}$	492	$2 \cdot 10^{-8}$
Cubic 1	70	$2.10 \cdot 10^{-5}$	1688	$3.38 \cdot 10^{-9}$
Cubic 2	79	$4.55 \cdot 10^{-5}$	1142	$9.82 \cdot 10^{-9}$
Cubic 3	124	$3.10 \cdot 10^{-6}$	1245	$1.66 \cdot 10^{-9}$
Cubic 4	141	$2.44 \cdot 10^{-6}$	993	$2.30 \cdot 10^{-9}$
Degree six	104	$9.94 \cdot 10^{-6}$	1083	$1.09 \cdot 10^{-8}$

The propagated profile has $B' \le 3$. The degree-six row additionally certifies $B/(1-m^2) > .5387$ and $3 - B' > .1165$. These are inherited completed records from the supplied certificate archives; the present consolidation changes neither their input functions nor their verification status.

B.6 Exact Profile Coefficients

The twenty coefficient vectors below specify the profiles B used in Lemma 2 and the profile positivity, closure, and base verifications in Appendix B. They are exact input data for those checks, rather than additional positivity claims. Let T_k denote the Chebyshev polynomial of the first kind, defined by

$$T_k(x) = \cos(k \arccos x), \qquad x \in [-1, 1].$$

Equivalently, $T_0 = 1$, $T_1 = m$, and $T_{j+1} = 2mT_j - T_{j-1}$. All profiles have the form

$$B(m) = (1-m^2)\left[1 + \frac{1}{D}\sum_{k=1}^{d} c_k\big(T_k(m) - T_k(0)\big)\right]. \tag{76}$$

Thus $B(0) = 1$ and $B(\pm 1) = 0$ are exact. The vector $\boldsymbol{c} = (c_1, \dots, c_d)$ is specified below. All integers below are literal coefficients, not rounded approximations. The entries below are also stored as JSON or Python integers in the accompanying checkers; the paper and program inputs agree term by term.

The family labels and bands refer to Tables 1. The following 14 legacy vectors correspond to Families I and II in the first 14 rows of the table and use denominator $D = 10^8$. The degree d below is the highest Chebyshev index; the degree of B itself is at most $d + 2$.

Family I, row 1: $s \in [33/40, 21/25]$**.**

$\boldsymbol{c} = (160139250,\ 100457153,\ 62927176,\ 43202696,\ 28512243,$
$19865464,\ 13042491,\ 8935333,\ 5620359,\ 3645946,$
$2097464,\ 1207376,\ 595378,\ 268798,\ 96620,$
$26219).$

Family I, row 2: $s \in [161/200, 33/40]$**.**

$\boldsymbol{c} = (157084175,\ 98312814,\ 61427917,\ 42184669,\ 27850902,$
$19421170,\ 12779784,\ 8765596,\ 5542051,\ 3596204,$
$2092262,\ 1203452,\ 608107,\ 271477,\ 104666,$
$25107).$

Family I, row 3: $s \in [77/100, 161/200]$**.**

$\boldsymbol{c} = (153593288,\ 96067166,\ 59851830,\ 41194618,\ 27191418,$
$18988994,\ 12488953,\ 8571083,\ 5418795,\ 3509820,$
$2054925,\ 1182354,\ 613867,\ 274916,\ 118576,$
$25792).$

Family I, row 4: $s \in [7/10, 77/100]$**.**

$\boldsymbol{c} = (137619871,\ 84301012,\ 51251990,\ 35183163,\ 22970610,$
$16100672,\ 10568346,\ 7339654,\ 4678933,\ 3074810,$
$1829431,\ 1064678,\ 572226,\ 252559,\ 117145,$
$21956).$

Family I, row 5: $s \in [3/5, 7/10]$**.**

$$\boldsymbol{c} = (116656053,\ 69165366,\ 39255588,\ 26485803,\ 15955995,\ 11047529,\ 6430919,\ 4522099,\ 2361990,\ 1644013,\ 679181,\ 488775,\ 126856,\ 90983).$$

Family I, row 6: $s \in [1/2, 3/5]$**.**

$$\boldsymbol{c} = (95240922,\ 55482694,\ 28843340,\ 19601809,\ 10780575,\ 7643742,\ 3906107,\ 2970335,\ 1255483,\ 1053185,\ 300637,\ 329838,\ 34543,\ 65320).$$

Family I, row 7: $s \in [49/100, 51/100]$**.**

$$\boldsymbol{c} = (85746073,\ 49947990,\ 23884249,\ 16079563,\ 7961122,\ 5609598,\ 2439830,\ 1870460,\ 578021,\ 532410,\ 82328,\ 99720).$$

Family I, row 8: $s \in [19/50, 49/100]$**.**

$$\boldsymbol{c} = (78358759,\ 47808638,\ 24385108,\ 17964418,\ 10230867,\ 7811849,\ 4468688,\ 3528163,\ 1950295,\ 1508444,\ 827848,\ 586202,\ 332685,\ 167915,\ 132492,\ 40563,\ 20963,\ 15411).$$

Family I, row 9: $s \in [33/100, 19/50]$**.**

$$\boldsymbol{c} = (67137548,\ 43428962,\ 20582854,\ 15963827,\ 8554789,\ 6923879,\ 3727730,\ 3131778,\ 1627266,\ 1344281,\ 715678,\ 511560,\ 299498,\ 141970,\ 133932,\ 25480,\ 24563,\ 14721).$$

Family I, row 10: $s \in [7/25, 33/100]$**.**

$$\boldsymbol{c} = (60219255,\ 38609319,\ 18075339,\ 14357130,\ 7079990,\ 6742582,\ 2580587,\ 3632783,\ 629030,\ 2060879,\ -91110,\ 1149695,\ -261874,\ 608553,\ -231468,\ 281628,\ -93327,\ 53531).$$

Family II, row 1: $s \in [0.24, 0.28]$**.**

$$\boldsymbol{c} = (53373236,\ 34516508,\ 15649519,\ 11528635,\ 6822949,\ 4278991,\ 3268753,\ 1577244,\ 1591729,\ 445825,\ 845589,\ -38490,\ 428957,\ -51625,\ 83483,\ 25100).$$

Family II, row 2: $s \in [0.21, 0.24]$**.**

$$\begin{aligned}\boldsymbol{c} = (&51666362,\ 32363105,\ 14987446,\ 12185851,\ 5358589,\\ &6218006,\ 1421970,\ 3795009,\ -239956,\ 2491463,\\ &-736110,\ 1544389,\ -653020,\ 856229,\ -413254,\\ &372154,\ -130795,\ 62417).\end{aligned}$$

Family II, row 3: $s \in [0.195, 0.21]$**.**

$$\begin{aligned}\boldsymbol{c} = (&50572922,\ 30791583,\ 16697765,\ 9601919,\ 8739444,\\ &2797979,\ 5216855,\ 541682,\ 3014574,\ -130499,\\ &1792873,\ -462582,\ 1172387,\ -475870,\ 632733,\\ &-263189,\ 265561,\ -102479,\ 62787,\ -5945).\end{aligned}$$

Family II, row 4: $s \in [0.19, 0.195]$**.**

$$\begin{aligned}\boldsymbol{c} = (&48487558,\ 33828457,\ 14640442,\ 13100870,\ 6222007,\\ &6486074,\ 2481412,\ 4038749,\ 395028,\ 2900459,\\ &-471490,\ 1959566,\ -616834,\ 1301490,\ -637890,\\ &889242,\ -481974,\ 476285,\ -238983,\ 184542,\\ &-66357,\ 29128).\end{aligned}$$

Family II, row 5: $s \in [0.18888, 0.19]$**.** Use $D = 10^{12}$.

$$\begin{aligned}\boldsymbol{c} = (&482020183983,\ 337104382572,\ 145039399217,\ 130252484468,\\ &61190442943,\ 64453835566,\ 24087988247,\ 40129151632,\\ &3413194010,\ 28890030198,\ -5103741617,\ 19567819347,\\ &-6439984820,\ 13046230167,\ -6582144158,\ 8961290804,\\ &-4945810843,\ 4798890956,\ -2434790762,\ 1848961974,\\ &-668278325,\ 286665359).\end{aligned}$$

For Families III and IV, we take $D = 10^{14}$.

Family III, row 1: $0.982 \le \rho \le 0.984$**.**

$$\begin{aligned}\boldsymbol{c} = (&47644271864007,\ 33995684173429,\ 15657040364901,\ 12523544832841,\\ &8467102365829,\ 5112416999675,\ 5102089620067,\ 2392020591900,\\ &2987874369747,\ 1155433625320,\ 1960006777408,\ 233441806477,\\ &1506225656211,\ -183248908793,\ 1006413858042,\ -235122027226,\\ &669083149185,\ -272212741368,\ 451630065618,\ -203378436771,\\ &232574882200,\ -84266653080,\ 67651277258,\ -532545928,\\ &174416990,\ 7184542494).\end{aligned}$$

Family III, row 2: $0.984 \le \rho \le 0.9846$**.**

$\boldsymbol{c}$ = (45271991367888, 37132651603678, 13793614711484, 15402351928700,
6832712687744, 7677034075156, 3610813706786, 4668505588720,
1594438840762, 3174060943885, 648743168296, 2024184919880,
263914686671, 1399277063494, −129715862099, 1091492364842,
− 281463442989, 759866792115, −281451317515, 543094274745,
− 278055936870, 388409753684, −219648169972, 230238132236,
− 108771377963, 81132590884, −21381093190, 15617820421,
− 2578303965, 3127021143).

Family III, row 3: $0.9846 \le \rho \le 0.9849$**.**

$\boldsymbol{c}$ = (51507244653755, 32816289575470, 16942052548911, 14745410334434,
6741051855281, 9195156330596, 2423823047144, 6257381741283,
782638620794, 4121025508411, 346061686665, 2480023109960,
285245268939, 1510851624309, 100380231547, 1026464181300,
− 62056659133, 770075873400, −263759804043, 731703047160,
− 457480121426, 702012283395, −519162594889, 606292806516,
− 455434823481, 434464733052, −302560175490, 240272978745,
− 139019641160, 80291783252, −27726687041, 8361128289).

Family III, row 4: $0.9849 \le \rho \le 0.985$**.**

$\boldsymbol{c}$ = (46621680063512, 36754857008954, 14047612199460, 16337746224504,
6230096071388, 8823221858899, 3269640466866, 5219798279544,
1732752747559, 3363170084137, 851887286931, 2182941944697,
410816357784, 1491208551425, 52142831698, 1094066625896,
− 116976308364, 774144459083, −196934189206, 584002382925,
− 243511915265, 448364362484, −262257478250, 376315180053,
− 272816214111, 304645535053, −220924436826, 195265178575,
− 118096566390, 72403084280, −25583195982, 8003290368).

Family IV, row 1: $0.985 \le \rho \le 0.9855$**.**

$\boldsymbol{c}$ = (53048501957628, 32229645434094, 19306730589981, 13438645504224,
9662367477315, 7566145098639, 5108650142030, 5081766185059,
2574402327257, 3776526117926, 1157642392365, 2826071763493,
406083320023, 2232416553896, −157651528226, 1867386287784,
− 457365985989, 1541005397700, −630855216601, 1339557363085,
− 744085032804, 1167424396570, −770996007417, 1011720085242,
− 740728996314, 850336799596, −657750535499, 687323187227,
− 539893358852, 511887196802, −391822356917, 337704780781,
− 238617756326, 172326889731, −95384858988, 49093476875,
− 16055620848, 4403332518).

C Certificates for Theorem 2

This appendix contains the exact kernel and profile data and verifies the four hypotheses of Proposition 1 for the five $\alpha = 13/25$ bands. Subsection C.1 provides the coefficients and kernel-positivity checks; C.2

derives the endpoint factorizations and verifies the pair inequality and kernel concavity; C.3 certifies the profile, closure, base, and gate conditions; and C.4 records the separate Hellinger-tail scalar certificate used in Theorem 2.

C.1 Exact Kernel and Profile Coefficients

The five bands are (27). All coefficients below are exact rationals in the raw entropy coordinate, with no rescaling. For $p \in I_i$ use

$$F(t) = t^{13/25} + (1-t)^{13/25} - 1, \qquad c = F(p), \qquad X(t) = F(t)^{12/13},$$
$$G_i(t) = F(t)\frac{2tP_i(F(t)) + X(t)N_i(F(t))}{P_i(F(t)) + X(t)N_i(F(t))}, \tag{77}$$
$$B_i(m) = (1-m^2)\sum_{j=0}^{6} b_{ij}m^j.$$

Here $N_i(z) = \sum_{j=0}^{6} n_{ij}z^j$ and $P_i(z) = \sum_{j=0}^{2} p_{ij}z^j$. All polynomial coefficients are in the raw variable $z = F(t)$, with no hidden rescaling. Every displayed terminating decimal is an exact rational number. In particular, $P_i(0) = 1$ and $d_i = N_i(0)$.

Table 2: Exact interface parameters.

band	d_i	ϕ_i	η_i	κ_i
I_1	1.38358473	0.92116540	1.19559623	1.01024700
I_2	1.34496105	0.86355952	1.16936066	0.96738500
I_3	1.33201793	0.83477456	1.14001962	0.95628000
I_4	1.32004896	0.80128678	1.20888682	0.95719200
I_5	1.31213681	0.77409622	1.21150798	0.96115200

Table 3: Exact coefficients n_{ij} of $N_i(z)$.

j	I_1	I_2	I_3	I_4	I_5
0	1.38358473	1.34496105	1.33201793	1.32004896	1.31213681
1	2.75210049	5.05044187	4.89730666	2.58981114	1.10521627
2	-6.35367220	-8.12600935	-3.76196510	1.07156821	-3.66052671
3	-27.61650993	-51.80420275	-81.28479594	-80.21298237	-24.82588574
4	4.17825002	76.94145017	178.79212563	185.35209941	13.51369992
5	72.94264032	-39.25314872	-213.31428588	-232.79481503	40.95278517
6	-99.22250911	-30.08816056	89.60268134	110.61077433	-62.67060986

Table 4: Exact coefficients p_{ij} of $P_i(z)$.

j	I_1	I_2	I_3	I_4	I_5
0	1.00000000	1.00000000	1.00000000	1.00000000	1.00000000
1	3.47187799	5.54977396	5.64986921	3.71371121	1.98159682
2	-14.28876797	-19.25266764	-19.25266764	-14.57643745	-10.79486261

The identities $G_i(0) = G_i(1) = 0$, $G_i(t) + G_i(1-t) = 2F(t)$, and $G_i'(0+) = d_i$ follow directly from (77). Every denominator is positive on $[0,1]$: one has $0 \le F(t) \le 2^{12/25} - 1 < 79/200$, where the final bound is the exact integer comparison $(279/200)^{25} > 2^{12}$. For each of the ten polynomials N_i, P_i, all Bernstein coefficients on $[0, 79/200]$ are strictly positive. These finite rational checks, with their lower and upper bounds, are recorded in `theorem2_auxiliary_certificate_manifest.json`. In particular $G_i(t) > 0$ for $0 < t < 1$. No concavity assertion is needed for this positivity check.

Table 5: Exact profile-factor coefficients b_{ij}.

j	I_1	I_2	I_3	I_4	I_5
0	1.00000000	1.00000000	1.00000000	1.00000000	1.00000000
1	0.16326039	0.14675115	0.15783272	0.17662864	0.20405980
2	0.06471946	0.10556021	0.11849323	0.13630788	0.15368011
3	0.04944774	0.05802038	0.05922484	0.04721638	0.04797380
4	0.13207191	0.04063015	0.04217959	0.03948286	0.03714184
5	0	0.04636980	0.04939686	0.09242399	0.12495400
6	0	0.11008254	0.11874468	0.10140824	0.09733462

C.2 Verification of the Pair Inequality

This subsection verifies the pair inequality (12) and the associated concavity requirement for the five $\alpha = 13/25$ kernels, as needed to apply Proposition 1 in the proof of Theorem 2. The kernel and profile coefficients are listed in appendix C.1; the closure, base, and gate conditions are checked in appendix C.3. The same pair verification procedure is applied to all five bands.

Endpoint factorizations. The following are algebraic identities for the rational exponents $13/25$ and $12/13$. They remove vanishing endpoint factors from the left and right pair residuals analyzed below; they are not additional standalone numerical inequalities.

The generic rational-exponent construction specializes as follows. For an endpoint coordinate $0 \le y < 1$, put

$$q_y = (1 - y^{25})^{1/25}, \qquad E(q) = \frac{\sum_{j=0}^{12} q^j}{\sum_{j=0}^{24} q^j}, \qquad \mathcal{B}(y) = 1 - y^{12}E(q_y), \qquad r_y = \mathcal{B}(y)^{1/13}.$$

The geometric-sum identity gives

$$F(y^{25}) = y^{13}\mathcal{B}(y), \qquad X(y^{25}) = y^{12}r_y^{12}.$$

Write $z_y = y^{13}\mathcal{B}(y)$ and evaluate N, P at z_y. Define

$$\mathcal{D}(y) = P(z_y) + y^{12}r_y^{12}N(z_y), \qquad \mathcal{A}(y) = \frac{r_y^{25}N(z_y) + 2y^{13}\mathcal{B}(y)P(z_y)}{\mathcal{D}(y)},$$

$$\mathcal{R}(y) = 2\mathcal{B}(y) - y^{12}\mathcal{A}(y), \qquad C_0(z) = \frac{dP(z) - N(z)}{z}, \qquad J_0(r) = \frac{\sum_{j=0}^{24} r^j}{\sum_{j=0}^{12} r^j}.$$

C_0 is a polynomial since $N(0) = dP(0) = d$. A second exact factorization is

$$\mathcal{M}(y) = \frac{y\mathcal{B}(y)C_0(z_y) + N(z_y)E(q_y)J_0(r_y) + dr_y^{12}N(z_y) - 2y\mathcal{B}(y)P(z_y)}{\mathcal{D}(y)}.$$

Consequently

$$G(y^{25}) = y^{25}\mathcal{A}(y) = y^{25}[d - y^{12}\mathcal{M}(y)], \qquad G(1 - y^{25}) = y^{13}\mathcal{R}(y). \tag{78}$$

In particular $\mathcal{A}(0) = d$, $\mathcal{M}(0) = d(1 + d)$ and $\mathcal{R}(0) = 2$. All explicit powers of y are nonnegative integers. If $v_p = p^{1/25}$, the entropy normalization is especially simple:

$$\frac{c^2}{p} = v_p\mathcal{B}(v_p)^2. \tag{79}$$

Thus the endpoint expressions require only positive 25th and 13th roots, together with rational operations.

For completeness, the diagonal factors are removed as follows. Put $P_{25}(r) = r^{25}$, $z_0 = 1 - a^{25}$,

$$w = (a^{25} + pz_0)^{1/25}, \qquad t = (1 - pz_0)^{1/25}, \qquad q = 1 - p.$$

For a function h, with divided differences extended continuously at repeated nodes, set

$$T_h = \frac{h[a,1]}{P_{25}[a,1]}, \qquad L_h(i) = \frac{h[a,i,1] - h[a,i]P_{25}[a,i,1]/P_{25}[a,i]}{P_{25}[a,1]P_{25}[i,1]}.$$

In the left chart $(u,v) = ((ab)^{25}, b^{25})$, use

$$h(r) = r^{37}\mathcal{M}(br), \qquad T_0 = \frac{q}{2}(L_h(w) + L_h(t)), \qquad T = T_h,$$

$$S_L = \phi[d - b^{12}\mathcal{M}(b)] + (1-\phi)a^{25}[d - (ab)^{12}\mathcal{M}(ab)].$$

The exact normalized left residual is

$$Q_L = T_0[S_L + \eta p b^{12} z_0^2 T_0] - \kappa\frac{c^2}{p}\frac{b^{12}T^2}{d^2}. \tag{80}$$

In the right chart $(u,v) = (1 - b^{25}, 1 - (ab)^{25})$, use

$$h(r) = r^{13}\mathcal{R}(br), \qquad T_0 = -\frac{q}{2}(L_h(w) + L_h(t)), \qquad T = T_h,$$

$$S_R = \phi a^{13}\mathcal{R}(ab) + (1-\phi)\mathcal{R}(b).$$

The normalized right residual is

$$Q_R = T_0[S_R + \eta p z_0^2 T_0] - \kappa\frac{c^2}{p}(b^{12} + T/d)^2. \tag{81}$$

For the common residual $\mathcal{P}_p$ the identities are

$$\mathcal{P}_p((ab)^{25}, b^{25}) = pb^{62}z_0^2 Q_L, \qquad \mathcal{P}_p(1 - b^{25}, 1 - (ab)^{25}) = pb^{26}z_0^2 Q_R.$$

The expressions Q_L, Q_R extend continuously to $a = 1$ and $b = 0$; those boundaries are not discarded.

The three charts and their complete coverage are described next.

Pair-region verification and concavity. All five kernels use (10).

For the residual $\mathcal{P} = D(S + \eta D) - \kappa c^2 W^2$ of (12), the three charts test

chart	(u,v)	normalized residual
left	$((ab)^{25}, b^{25})$	$\mathcal{P}/[pb^{62}(1-a^{25})^2]$
right	$(1 - b^{25}, 1 - (ab)^{25})$	$\mathcal{P}/[pb^{26}(1-a^{25})^2]$
cross	$(l^{25}, 1 - a^{25})$	$\mathcal{P}/p$

with $0 \le a \le 1$, $0 \le b \le .99$ in the side charts and $0 \le l, a \le .95$ in the cross chart. Since $.99^{25} > 3/4$ and $1/4 < .95^{25} < 3/10$, these charts cover every ordered pair; cross-chart pairs have $v - u > 2/5$. The side expressions extend continuously to $a = 1$ and $b = 0$ by the endpoint factorizations above. Their strictly positive diagonal limits imply

$$-\frac{p(1-p)}{2}G(t)G''(t) - \kappa c^2(1 - G'(t)/d)^2 > 0.$$

As $G > 0$, this proves concavity on both side interiors, whose union is $(0,1)$; only then is concavity used to infer $D \ge 0$.

Divided differences and directed roots. Divided differences are continuously extended at repeated nodes. If $y^h = x > 0$, put $S_h(u,v) = \sum_{j=0}^{h-1} u^j v^{h-1-j}$ and $T_{h-2}(u,v,w) = \sum_{i+j+k=h-2} u^i v^j w^k$. The exact identities

$$y_{ij} = \frac{x_{ij}}{S_h(y_i, y_j)}, \qquad y_{012} = \frac{x_{012} - T_{h-2}(y_0, y_1, y_2) y_{01} y_{12}}{S_h(y_0, y_2)}$$

propagate interval enclosures through positive 13th and 25th roots; products use $(fg)_{012} = f_0 g_{012} + f_{01} g_{12} + f_{012} g_2$. Root endpoints are checked by exact integer comparisons of dyadic powers. Thus library root approximations serve only as proposals.

Interior derivative bounds. Interval automatic differentiation encloses $G^{(k)}/k!$ by C_k, $0 \le k \le 3$, throughout a compact interval containing all nodes. The Hermite–Genocchi identity

$$G[x_0, \dots, x_k] = \int_{\Delta_k} G^{(k)}\left(\sum_i \lambda_i x_i\right) d\lambda, \qquad \int_{\Delta_k} 1 = 1/k!,$$

and $\int_{\Delta_k} \lambda_i = 1/(k+1)!$ give the bounds

$$G[x_0, \dots, x_k] \in C_k, \qquad \partial_j G[x_0, \dots, x_k] \in \sum_i C_{k+1} V_{ij} \quad (k = 1, 2),$$

where V_{ij} encloses $\partial_j x_i$. Each product is enclosed separately, including at repeated nodes. These bounds are intersected with the algebraic ones. The derivative table uses complete dyadic cells $[k/8192, (k+1)/8192]$; every interior query is enclosed by a verified union of cells, never by sampled derivatives. For $z = v - u$, the equivalent residual

$$\frac{\mathcal{P}}{pz^2} = T_0(S + \eta p z^2 T_0) - \kappa \frac{c^2}{p}\left(1 - \frac{G[u,v]}{d}\right)^2, \qquad T_0 = -\frac{1-p}{2}\{G[u, u+pz, v] + G[u, v-pz, v]\}$$

allows the same bounds in each chart. It is used only for interior physical nodes; endpoint boxes retain the algebraic factorizations.

Acceptance and coverage. All arithmetic uses outward binary80 intervals, with 64-bit significands, explicit outward enlargement, and no unsafe fast-math or fused-multiply-add contraction. A box passes only by positive direct enclosure or a positive center bound minus full-box gradient variation. All other boxes are subdivided, retaining shared boundaries. Arithmetic failures and unresolved boxes never count as certificates. Exact continuation preserves the partition into accepted leaves and the pending stack: candidate, domain, source, binary and predecessor hashes are checked, and pending endpoints are copied as representable hexadecimal dyadics. Completion requires every pending box to pass. The final records are

Band	left leaves	right leaves	cross leaves
I_1	21420	34315	20519
I_2	24883	34019	36136
I_3	20154	31079	19751
I_4	20759	26276	17125
I_5	21455	30789	12036

All fifteen records are complete and hash-bound to the displayed data. Counts include certified predecessor leaves; unavailable observed minima are not reconstructed. The independent full-domain runner is specified in Appendix D.

C.3 Verification of the Closure, Base, and Gate Conditions

This subsection supplies the non-pair conditions of Proposition 1: the profile normalization and positivity, closure (13), the three-bit base inequalities (14), and gate (15). The pair condition is treated in appendix C.2.

For each band suppress the subscript i and set

$$b_4(m) = \min\{3/8, 9/16 - |m|/2, 1 - |m|\}, \qquad 0 \le x \le b_4(m),$$

$$A = \frac{2B(m) - B(m+x) - B(m-x)}{2x^2}, \qquad C = \frac{B(m+x) - B(m-x)}{2x}, \qquad J = 1 - \frac{2c}{d}C,$$

with polynomial continuations at $x = 0$. Define the closure residual

$$C = \kappa J^2 - A\big[B(m) + (2\phi - 1)xC + (\eta - 1)x^2 A\big].$$

All five profiles satisfy $B \ge 0$, $B(0) = 1$ and $B(\pm 1) = 0$. The following table lists strict rational lower bounds certified uniformly over the whole corresponding noise band and influence domain. The base residual is $\mathbb{E}G((1 + T_\rho g)/2) - cB(\mathbb{E}g)$. In addition, throughout all five bands,

Table 6: Exact auxiliary margins and subdivision sizes.

band	C	nontrivial base residual	closure leaves	base slabs
I_1	0.0046	$4 \cdot 10^{-7}$	12	115
I_2	0.0007	10^{-7}	14	483
I_3	0.00034	10^{-6}	15	256
I_4	0.00052	$5 \cdot 10^{-6}$	15	256
I_5	0.00178	$2 \cdot 10^{-5}$	11	64

$$J > 0.79, \qquad 1 - \frac{\max(\phi, 1-\phi)}{\eta} - \frac{2c}{d}\sqrt{\frac{\kappa}{\eta}} > 0.22. \tag{82}$$

The nontrivial base bound applies to fifteen increasing three-bit truth tables. The remaining five tables—two constants and three dictators— satisfy the base inequality as symbolic equalities. This covers all sources with at most three active coordinates, including dummy variables.

The verifier makes every acceptance decision using exact integers or rational numbers. It forms A, C algebraically, splits the influence domain at $m = -7/8, -3/8, 3/8, 7/8$, maps each piece by $m = \ell + (h - \ell)u$, $x = b_4(m)v$, and checks the tensor Bernstein coefficients on $[0, 1]^2$, subdividing by de Casteljau when necessary. Since $J > 0$, the closure expression is monotone in c for each fixed (m, x), with direction determined by the sign of C; only the two outward-enclosed noise endpoints need to be checked. Bases are checked from all twenty truth tables using their exact degree-three probability polynomials in p. The powers $13/25$ and $12/13$ use directed integer roots at denominator 2^{192}. No sampled sign is accepted as a proof.

C.4 Hellinger Tail Certificate

The scalar root comparison (28), used together with Lemma 4 and Theorem 3 for the remaining near-one correlation interval, is checked with directed integer roots at $s_0^2 = 999999/250000000000$. It is distinct from the pair, closure, base, and gate certificates.

D Computational Verification and Reproducibility

This appendix identifies the computational checks used in the proofs of Theorems 1 and 2. In particular, A.5 records the scalar positivity tests for Lemmas 8, 9, and 11; Appendix B supports Lemma 2;

Appendix C supports the $\alpha = 13/25$ application of Proposition 1 and the Hellinger tail. These checks are separate from the externally cited Hellinger contraction theorem.

The finite computations in this paper are restricted to strict positivity of explicit polynomial or pair-residual functions over specified compact domains. The accompanying source package contains exact rational Bernstein verifiers, directed integer-root interval arithmetic, and outward-rounded binary80 C++ pair-domain certifiers.

For the square-root comparison, the listed checks comprise six groups of scalar semigroup inequalities in Appendix A; profile positivity and closure, and fifteen nontrivial three-bit base cases for each of the twenty parameter rows; and the pair charts in Appendix B. The Family-I–II pair requirements are grouped in appendix B.3, with the last Family-II band propagated by appendix B.4; the Family-III–IV records are in appendix B.5. The original source package and completed-run receipts are needed to reproduce or independently audit these reported results. The strengthened band through $\rho = 1971/2000$ is included.

For the order $13/25$ comparison, the five exact auxiliary certificates, the ten polynomial kernel-positivity conditions, the Hellinger scalar tail inequality, and the fifteen rational-power pair charts are treated separately; their verification status and reproduction instructions are recorded in the accompanying report. The external Hellinger contraction inequality is not a computational consequence of the supplied code.

A fresh complete verification can be initiated by running

```
python3 verify_all.py all --jobs 3
```

This command requires Python 3.10 or later, SymPy, and a C++17 compiler supporting the checked binary80 long-double format. An unresolved region or interrupted program is not a certificate. Elementary polynomial identities and rational interval unions are proved algebraically and do not require numerical verification.

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Generative-AI use disclosure: ChatGPT was used to explore proof strategies, develop and refine mathematical arguments and exposition, and assist in preparing computational verification procedures. Building on the dimension-induction approach developed in [6], the author used ChatGPT to investigate various modifications of the auxiliary kernels and profiles introduced therein, with the aim of proving Conjecture 1. Through these iterative exchanges, the specific functional forms adopted in this paper were identified and refined. All mathematical arguments presented in the final manuscript were independently verified by the author, who assumes full responsibility for the correctness and content of the paper.

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